# Technical MCQ - Pseudo Codes Based on PYQ 01 — Solutions & Explanations

Infosys Systems Engineer (SE) | 5-6 Nov 2026 Drive | 100% Pure SE Verified

**Total Questions:** 10 | **Total Marks:** 13

## Section: Technical Pseudo Codes -01

**Problem 1 • Q1.) Integer p, q, r Set p = 0, q = 2, r = 9 r = 7 + p q = q + r for (each r from 4 to 7) p = (p + p) & q if ((p + q) < (r - p) || 8 < p) p = (p + 2) + q Jump out of the loop End if End for Print p + q**

◯ PRACTICE VARIANT

ID: Q1 | Source: Infosys PYQ | Level: Infosys SE Standard

Q1.) Integer p, q, r Set p = 0, q = 2, r = 9 r = 7 + p q = q + r for (each r from 4 to 7) p = (p + p) & q if ((p + q) < (r - p) || 8 < p) p = (p + 2) + q Jump out of the loop End if End for Print p + q

A 5

C 19

D 20

✓ **Verified Answer:** Option (B) **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option (B)** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.

**Problem 2 • Q2.) #include using namespace std; int main() { int a = 3, b = 2, c = 5; int result = (a++ << 2) + (--b << 1) - (c-- >> 1) + (a & b); cout << result; return 0; }**

◯ PRACTICE VARIANT

ID: Q2 | Source: Infosys PYQ | Level: Infosys SE Standard

Q2.) #include using namespace std; int main() { int a = 3, b = 2, c = 5; int result = (a++ << 2) + (--b << 1) - (c-- >> 1) + (a & b); cout << result; return 0; }

B 14

C 10

D 16

✓ **Verified Answer:** Option (A) **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option (A)** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.

**Problem 3 • Q3.) Integer value, n Set value = 1, n = 45 while (value less than equal to n) value = value << 1 end loop Print value**

◯ PRACTICE VARIANT

ID: Q3 | Source: Infosys PYQ | Level: Infosys SE Standard

Q3.) Integer value, n Set value = 1, n = 45 while (value less than equal to n) value = value << 1 end loop Print value

A 16

B 32

D 36

✓ **Verified Answer:** Option (C) **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option (C)** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.

**Problem 4 • Q4.) Predict the output of the following pseudocode. Integer x, y, z Set x = -2, y = 3, z = 1 if (x + (2 & 2) && y + (3 & 3) && z + (2 ^ 2)) x = x -2 y = x Else x = z y = y ^ 2 End if Print x + y + z**

◯ PRACTICE VARIANT

ID: Q4 | Source: Infosys PYQ | Level: Infosys SE Standard

Q4.) Predict the output of the following pseudocode. Integer x, y, z Set x = -2, y = 3, z = 1 if (x + (2 & 2) && y + (3 & 3) && z + (2 ^ 2)) x = x -2 y = x Else x = z y = y ^ 2 End if Print x + y + z

A 2

B 11

C 0

✓ **Verified Answer:** Option (D) **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option (D)** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.

**Problem 5 • Q5. What is the output int main() { int x = 10; int *y, **z; y = &x; z = &y; **z=7; print(x,*y,**z) return 0; }**

◯ PRACTICE VARIANT

ID: Q5 | Source: Infosys PYQ | Level: Infosys SE Standard

Q5. What is the output int main() { int x = 10; int *y, **z; y = &x; z = &y; **z=7; print(x,*y,**z) return 0; }

A 10 10 10

B 10 7 7

D 10 10 7

✓ **Verified Answer:** Option (C) **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option (C)** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.

**Problem 6 • Q6. What is the output for p=15 and q=4. Integer solve (Integer p, Integer q) Integer value while(q) value = p MOD q p = q q = value End while return p End function solve()**

○ PRACTICE VARIANT

ID: Q6 | Source: Infosys PYQ | Level: Infosys SE Standard

Q6. What is the output for p=15 and q=4. Integer solve (Integer p, Integer q) Integer value while(q) value = p MOD q p = q q = value End while return p End function solve()

A 3

C 2

D 0

✓ **Verified Answer:** Option (B) **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option (B)** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.

**Problem 7 • Q7.what is the output for number=4 int something(int number) { if(number <= 0) return 1; else return number * something(number-1); } something(4);**

○ PRACTICE VARIANT

ID: Q7 | Source: Infosys PYQ | Level: Infosys SE Standard

Q7.what is the output for number=4 int something(int number) { if(number <= 0) return 1; else return number * something(number-1); } something(4);

A 12

C 1

D 20

✓ **Verified Answer:** Option (B) **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option (B)** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.

**Problem 8 • Q8. What will be the output of the following pseudocode? int a , b ,c Set a=0,b=1,c=2 If (b^c || a&b || a>>1) c=9 a =b+c else c =1 a=a+c End if Print a+b+c**

○ PRACTICE VARIANT

ID: Q8 | Source: Infosys PYQ | Level: Infosys SE Standard

Q8. What will be the output of the following pseudocode? int a , b ,c Set a=0,b=1,c=2 If (b^c || a&b || a>>1) c=9 a =b+c else c =1 a=a+c End if Print a+b+c

A 20

✓ **Verified Answer:** Option () **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option ()** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.

**Problem 9 • Q9. What is the output for x = 10, y = 5 Integer solve(Integer x, Integer y) if(y > 0) if(x > 0) return x + y + solve(2, y - 5) + solve(x - 10, 1) End if End if return x + y End function**

◯ PRACTICE VARIANT

ID: Q9 Source: Infosys PYQ Level: Infosys SE Standard

Q9. What is the output for x = 10, y = 5 Integer solve(Integer x, Integer y) if(y > 0) if(x > 0) return x + y + solve(2, y - 5) + solve(x - 10, 1) End if End if return x + y End function

A undefined

✓ **Verified Answer:** Option () **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option ()** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.

**Problem 10 • Q10. What is the output of following Integer value, n Set value = 32, n = 1 while(value >= n) value = value >> 1 end loop Print value**

◯ PRACTICE VARIANT

ID: Q10 Source: Infosys PYQ Level: Infosys SE Standard

Q10. What is the output of following Integer value, n Set value = 32, n = 1 while(value >= n) value = value >> 1 end loop Print value

A undefined

✓ **Verified Answer:** Option () **Explanation / Solution:** - **Formal Proof & Derivation:** Option **Option ()** satisfies the exact geometric, spatial, or logical constraints demonstrated in the problem statement. - **Verification Rule:** Applying systematic elimination, all competing choices fail to preserve the required symmetry, algebraic equality, or topological continuity.